Free-streaming DR vs LCDM (TT)

FS ΔN=0.05
FS ΔN=0.1
FS ΔN=0.2
FS ΔN=0.33

$\Delta C_\ell^{TT}/C_\ell^{TT}$ [%]

ℓ

Phase shift illustration (ΔN=0.2)
Expected shift ~ 0.89 deg

ΛCDM
FS ΔN=0.2

$\ell(\ell+1)C_\ell^{TT}/2\pi$ [μK^2]

ℓ

Free-streaming DR vs LCDM (EE)

FS ΔN=0.05
FS ΔN=0.1
FS ΔN=0.2
FS ΔN=0.33

$\Delta C_\ell^{EE}/C_\ell^{EE}$ [%]

ℓ

Fluid vs FS distinguishing: phase shift

Phase shift (degrees)
~CMB-S4 resolution
~Planck resolution

Phase shift [degrees]

ΔN_{fluid}